2024 AMC 10B Problem 23 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 23. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 23

Problem:

The Fibonacci numbers are defined by $F_1 = 1, F_2 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 3$. What is

$$\frac{F_2}{F_1} + \frac{F_4}{F_2} + \frac{F_6}{F_3} + \cdots + \frac{F_{20}}{F_{10}}?$$

Answer Choices:

A. 318

B. 319

C. 320

D. 321

E. 322

Solution:

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597, 2584, 4181, 6765, \ldots,$$

so the given sum is

$$\frac{1}{1}+\frac{3}{1}+\frac{8}{2}+\frac{21}{3}+\frac{55}{5}+\frac{144}{8}+\frac{377}{13}+\frac{987}{21}+\frac{2584}{34}+\frac{6765}{55}$$

which equals $1+3+4+7+11+18+29+47+76+123=\text{(B)}\boxed{319}$.

OR

The Fibonacci sequence starts out $1,1,2,3,5,8,13,21,34,55$, so the given sum starts out

$$\frac{1}{1}+\frac{3}{1}+\frac{8}{2}+\frac{21}{3}+\frac{55}{5}=1+3+4+7+11$$

It appears that these summands satisfy the same recurrence relation, namely

$$\frac{F_{2n}}{F_n}=\frac{F_{2(n-1)}}{F_{n-1}}+\frac{F_{2(n-2)}}{F_{n-2}}$$

With the initial conditions 1,3 instead of 1 , 1 , the sequence

$$(L_n)=\left(\frac{F_{2n}}{F_n}\right)$$

is known as the Lucas sequence. If the recurrence above is correct, then the required sum is

$$1+3+4+7+11+18+29+47+76+123=319$$

To prove the identity for (L_n) displayed above, recall Binet's formula, $F_n=\frac{1}{\sqrt{5}}\left(\phi^n-\psi^n\right)$, where $\phi=\frac{1+\sqrt{5}}{2}$ and $\psi=\frac{1-\sqrt{5}}{2}$ are the roots of the polynomial x^2-x-1. Then

$$L_n = \frac{F_{2n}}{F_n} = \frac{\phi^{2n} - \psi^{2n}}{\phi^n - \psi^n} = \phi^n + \psi^n.$$

Therefore

$$\begin{aligned} L_{n-1} + L_{n-2} &= \phi^{n-1} + \phi^{n-2} + \psi^{n-1} + \psi^{n-2} \\ &= \phi^{n-2}(\phi + 1) + \psi^{n-2}(\psi + 1) \\ &= \phi^{n-2} \cdot \phi^2 + \psi^{n-2} \cdot \psi^2 \\ &= \phi^n + \psi^n = L_n. \end{aligned}$$